Architecture Overview - Harborview Platform

This document describes the architecture of the Harborview Platform built for Brightwater Logistics on the Falcon engagement, as revised on 11 September 2026 for Phase 2. It covers the components, the data flow between them and the privacy rules that govern what leaves a user's laptop. It is the reference for engineers, the security reviewer and the client delivery team.

Components

Component	Technology	Responsibility
Local agent	Python service on 127.0.0.1	Reads the user's folders, extracts text, builds the local index, answers searches
Web frontend	React single-page app	Chat-style search and Ask; talks only to the local agent
Central service	Serverless functions behind an API gateway	Teams, public file metadata, relay for downloads, retrieval of public chunks
Central database	Postgres with a vector extension	File metadata and the stored text chunks of public files
Transfer relay	Object storage, 15 minute lifetime	Short-lived hand-off of a file between two laptops
Managed model service	Hosted language model	Tag generation and Ask answers

Data flow

- Search: the frontend sends the question to the local agent, which searches the user's own files first and then public metadata held centrally.
- Publish: when a file is made public, the agent uploads its metadata, tags and text chunks with embeddings to the central service.
- Ask: the agent embeds the question locally, sends only the query vector to central retrieval, and builds the answer from the returned chunks.
- Download: the owner's agent puts the file in the transfer relay under a random key; the requester fetches it once and the object is swept.

Privacy rules

- Private files are never uploaded: not the bytes, not the text, not the chunks.
- Public files have their text chunks stored centrally and readable by members of the same team. Publishing means the content is readable by teammates.
- Chunks are removed when a file is unpublished, deleted or expires after 45 days.
- Request bodies, queries, chunk text and file content are never written to logs.
- The frontend never calls the central service directly; every call goes through the local agent.

Owner of this document: Delivery lead (User 2). Reviewers: engineers and the Client delivery lead.